

Theo
SOULAT
BTS SIO 1 LM

TP Administration Réseau



Plan d'adressage :

VLAN 10 : Administration

Hôtes requis : 170

Masque : /24

Adresse réseau : 172.20.0.0

1^{er} adresse utilisable : 172.20.0.1

Dernier adresse utilisable : 172.20.0.254

Adresse de broadcast : 172.20.0.255

VLAN 20 : Equipes

Hôtes requis : 164

Masque : /24

Adresse réseau : 172.20.1.0

1^{er} adresse utilisable : 172.20.1.1

Dernier adresse utilisable : 172.20.1.254

Adresse de broadcast : 172.20.1.255

VLAN 30 : WIFI

Hôtes requis : 100

Masque : /25

Adresse réseau : 172.20.2.0

1^{er} adresse utilisable : 172.20.2.1

Dernier adresse utilisable : 172.20.2.126

Adresse de broadcast : 172.20.2.127

VLAN 40 : VIP-Press

Hôtes requis : 80

Masque : /25

Adresse réseau : 172.20.2.128

1^{er} adresse utilisable : 172.20.2.129

Dernier adresse utilisable : 172.20.2.254

Adresse de broadcast : 172.20.2.255

VLAN 50 : Cameras IP

Hôtes requis : 80

Masque : /25

Adresse réseau : 172.20.3.0

1^{er} adresse utilisable : 172.20.3.1

Dernier adresse utilisable : 172.20.3.126

Adresse de broadcast : 172.20.3.127

VLAN 60 : Fournisseurs

Hôtes requis : 44

Masque : /26

Adresse réseau : 172.20.3.128

1^{er} adresse utilisable : 172.20.3.129

Dernier adresse utilisable : 172.20.3.190

Adresse de broadcast : 172.20.3.191

VLAN 70 : Restaurant

Hôtes requis : 170

Masque : /27

Adresse réseau : 172.20.3.192

1^{er} adresse utilisable : 172.20.3.193

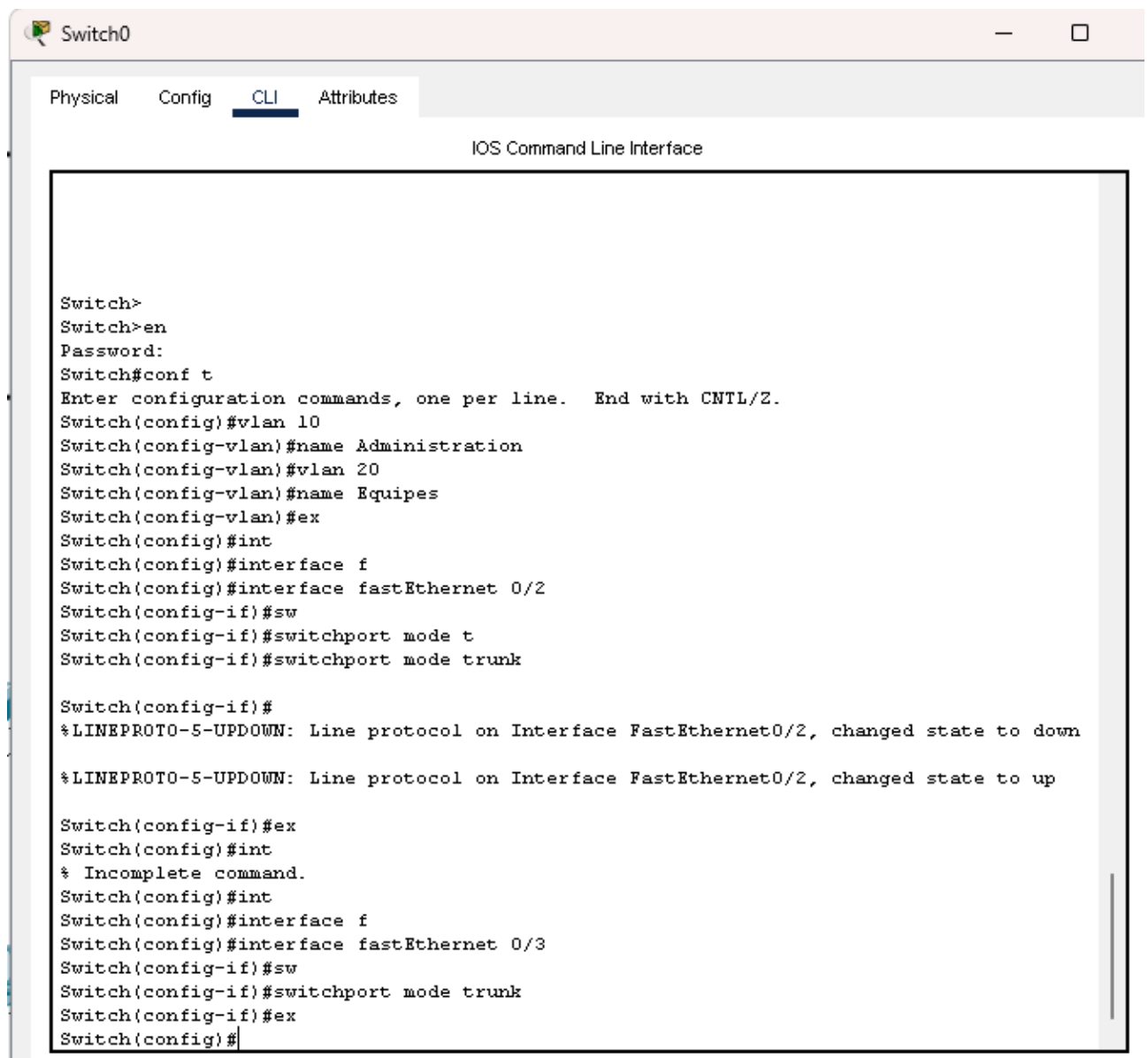
Dernier adresse utilisable : 172.20.3.222

Adresse de broadcast : 172.20.3.223

On configure un mot de passe (« admin ») sur tous les switches et routeur :

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#en
Switch(config)#en se
Switch(config)#enable secret admin
```

On configure les vlan sur le switch du VLAN Administration :



The screenshot shows a web-based configuration interface for a switch named 'Switch0'. The 'CLI' tab is selected, displaying the 'IOS Command Line Interface'. The terminal shows the following sequence of commands and responses:

```
Switch>
Switch>en
Password:
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Administration
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name Equipes
Switch(config-vlan)#ex
Switch(config)#int
Switch(config)#interface f
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#sw
Switch(config-if)#switchport mode t
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

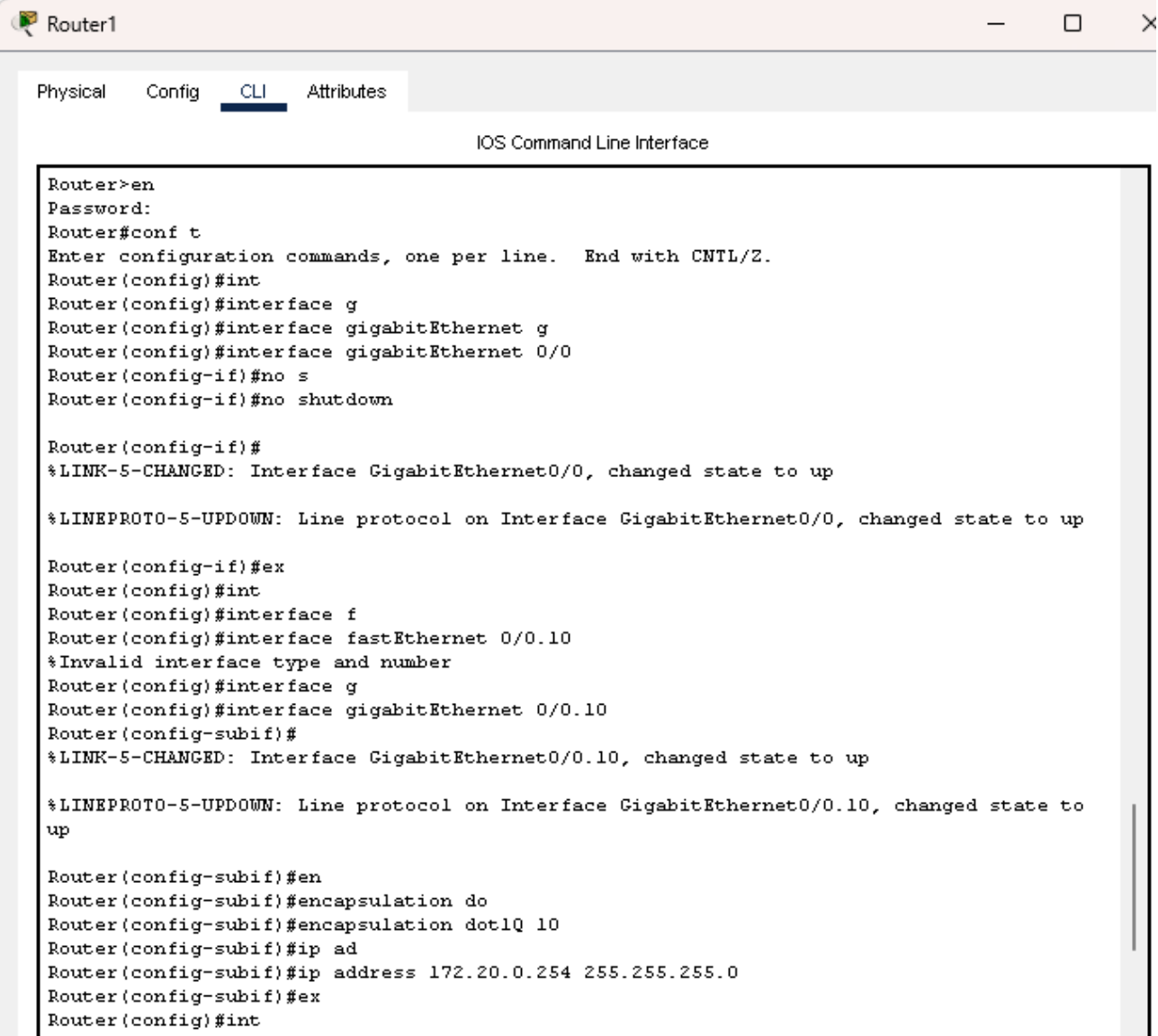
Switch(config-if)#ex
Switch(config)#int
% Incomplete command.
Switch(config)#int
Switch(config)#interface f
Switch(config)#interface fastEthernet 0/3
Switch(config-if)#sw
Switch(config-if)#switchport mode trunk
Switch(config-if)#ex
Switch(config)#
```

On configure les vlan sur le switch du VLAN Equipe :

```
Switch>en
Password:
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Administration
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name Equipes
Switch(config-vlan)#ex
Switch(config)#int
Switch(config)#interface f
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#sw
Switch(config-if)#switchport mode trun
Switch(config-if)#switchport mode trunk
Switch(config-if)#ex
Switch(config)#ex
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
Switch#
```

On configure le router qui est connecter au switch du vlan administration :



```
Router1
Physical Config CLI Attributes
IOS Command Line Interface

Router>en
Password:
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int
Router(config)#interface g
Router(config)#interface gigabitEthernet g
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#no s
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#ex
Router(config)#int
Router(config)#interface f
Router(config)#interface fastEthernet 0/0.10
%Invalid interface type and number
Router(config)#interface g
Router(config)#interface gigabitEthernet 0/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#en
Router(config-subif)#encapsulation do
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip ad
Router(config-subif)#ip address 172.20.0.254 255.255.255.0
Router(config-subif)#ex
Router(config)#int

-----
Router(config)#interface f
Router(config)#interface fastEthernet 0/0.10
%Invalid interface type and number
Router(config)#interface g
Router(config)#interface gigabitEthernet 0/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#en
Router(config-subif)#encapsulation do
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip ad
Router(config-subif)#ip address 172.20.0.254 255.255.255.0
Router(config-subif)#ex
Router(config)#int
Router(config)#interface g
Router(config)#interface gigabitEthernet 0/0.20
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#e
Router(config-subif)#e
Router(config-subif)#en
Router(config-subif)#encapsulation
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip ad
Router(config-subif)#ip address 172.20.1.254 255.255.255.0
Router(config-subif)#ex
Router(config)#
```

Configuration des routeurs :

billetterie :

```
interface GigabitEthernet0/0
 ip address 192.168.1.254 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/3/0
 ip address 200.200.200.6 255.255.255.252
 clock rate 2000000
!
```

magasin :

```
interface GigabitEthernet0/0
 ip address 192.168.2.254 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/3/0
 ip address 200.200.200.10 255.255.255.252
 clock rate 2000000
 shutdown
!
```

Routeur Principale :

```
interface Serial0/2/0
 ip address 200.200.200.5 255.255.255.252
 shutdown
!
interface Serial0/2/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/3/0
 ip address 200.200.200.1 255.255.255.252
 shutdown
!
interface Serial0/3/1
 ip address 200.200.200.9 255.255.255.252
 shutdown
!
```

Routeur VLAN :

```
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 172.20.0.254 255.255.255.0
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 172.20.1.254 255.255.255.0
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/3/0
 ip address 200.200.200.2 255.255.255.252
 clock rate 2000000
 shutdown
!
```

PING du PCO vers tout les autres poste :

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=2ms TTL=125
Reply from 192.168.1.1: bytes=32 time=2ms TTL=125
Reply from 192.168.1.1: bytes=32 time=3ms TTL=125
Reply from 192.168.1.1: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\>ping 172.20.0.1

Pinging 172.20.0.1 with 32 bytes of data:

Request timed out.
Reply from 172.20.0.1: bytes=32 time=33ms TTL=125
Reply from 172.20.0.1: bytes=32 time=26ms TTL=125
Reply from 172.20.0.1: bytes=32 time=2ms TTL=125

Ping statistics for 172.20.0.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 33ms, Average = 20ms

C:\>ping 172.20.1.1

Pinging 172.20.1.1 with 32 bytes of data:

Request timed out.
Reply from 172.20.1.1: bytes=32 time=3ms TTL=125
Reply from 172.20.1.1: bytes=32 time=2ms TTL=125
Reply from 172.20.1.1: bytes=32 time=2ms TTL=125

Ping statistics for 172.20.1.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\>
```

Schema Cisco Packet Tracer :

